

Enron Email Corpus – Week 2

Variables, distributions, and what they show

Data Mining Lab SoSe 26 – Task 2

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What Enron was

Enron was a large US energy and commodities trading company based in Houston. After a decade of rapid growth it collapsed in late 2001 in an accounting-fraud scandal, then among the largest US bankruptcies.

The mailboxes survived because the Federal Energy Regulatory Commission placed them in the public record during the investigation. Carnegie Mellon hosts the cleaned version we use.

Dataset: `cs.cmu.edu/~enron/` `enron_mail_20150507.tar.gz`

What we have

- About **517,000** email files
- From **150** mailbox owners
- Covering **1997 to 2002** (Enron's active years)
- Unpacked: roughly **2.6 GB** of plain text

A few hundred files carry broken dates and a handful of folders mix sent and received mail, so later plots are drawn after a light cleaning pass.

One file, one email

Every message is a plain-text file in the RFC-822 format. A block of headers is followed by the body.

```
Message-ID: <1326004.1075846655910.JavaMail.evans@thyme>  
Date:      Wed, 18 Apr 2001 04:08:00 -0700 (PDT)  
From:      mary.cook@enron.com  
To:        angela.davis@enron.com, ... , tana.jones@enron.com  
Subject:   Guaranty Distribution  
X-From:    Mary Cook  
X-Origin:  BAILEY-S  
X-Folder:  \Susan_Bailey_June2001\Notes Folders\All documents
```

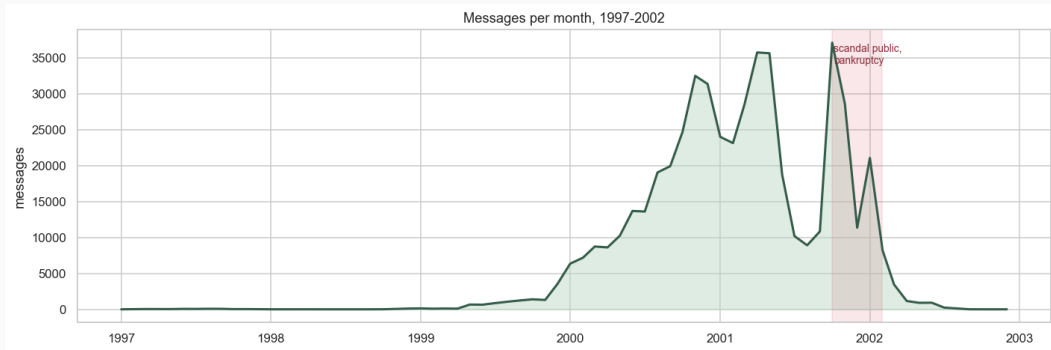
As a reminder, paralegals should distribute guaranties as stated ...

Variables

The fields below are kept from each message. X-* fields are Outlook-specific and tell us which mailbox the file came from.

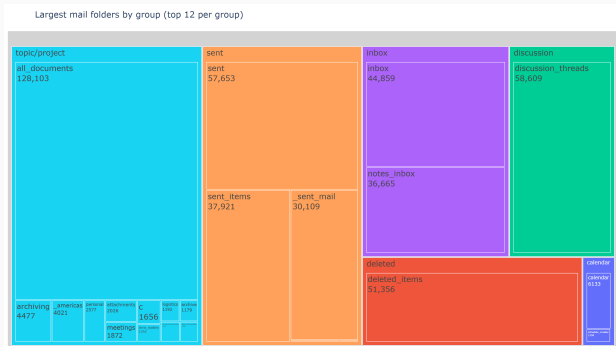
Field	Example value	Data type
Message-ID	<1326004...@thyme>	Nominal (unique string)
Date	18 Apr 2001 04:08 PDT	Datetime
From	mary.cook@enron.com	Nominal (categorical)
To	list of addresses	Nominal (list)
Subject	Guaranty Distribution	Text
X-From	Mary Cook	Text (display name)
X-Origin	BAILEY-S	Mailbox identifier
Message body	free text	Text (freeform)

Mail volume over time



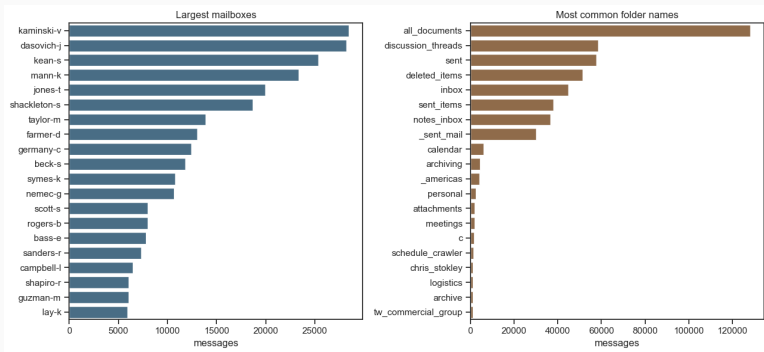
Traffic climbs through 2000 and peaks at **37,139** messages in October 2001, the same month the fraud became public.

Folder structure



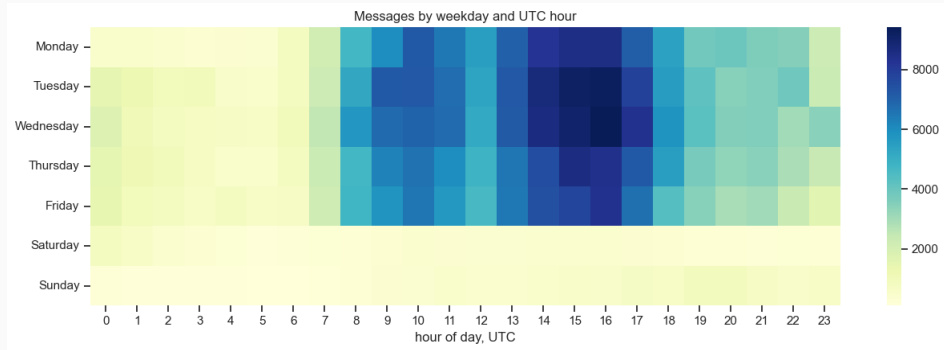
Folders fall into six broad groups. all_documents, sent, inbox and deleted_items dominate.

Largest mailboxes and folders



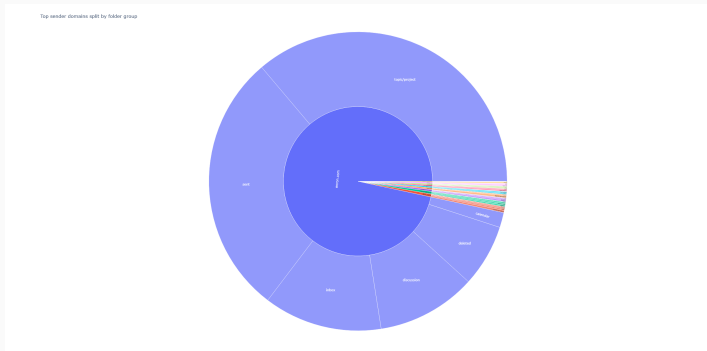
kaminski-v is the largest single mailbox (**28,465** files). The ten biggest mailboxes hold **37.7%** of the corpus.

Activity by weekday and hour



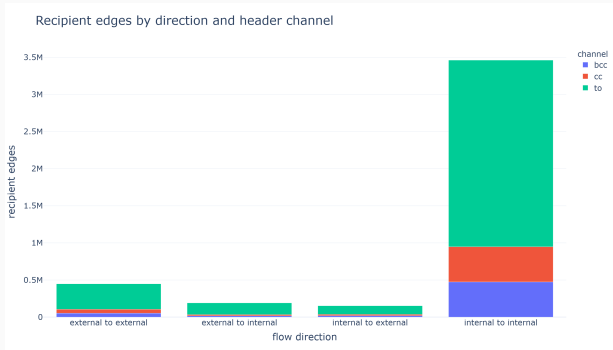
Weekend traffic is rare (**3.8%**). Weekday activity peaks in the afternoon UTC, with a visible lunch dip and a lighter Friday.

Sender domains



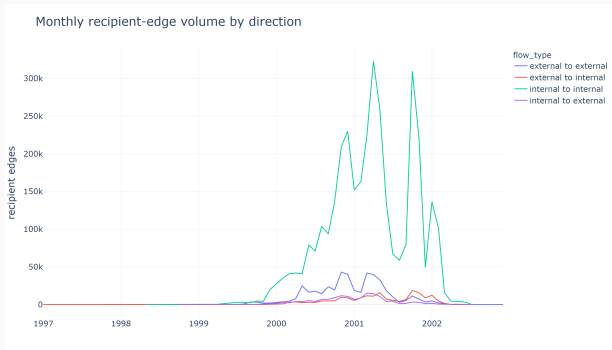
About **83%** of mail comes from an `enron.com` address. External domains together make up only a thin slice.

Recipient edges by direction



Each message expands into edges (one per To/Cc/Bcc). Internal-to-internal traffic is the large majority.

Edge volume over time



The internal-vs-external mix stays roughly stable across the company's history, even as overall volume rises and falls.

What people wrote about



Common English words sit on top of a clearly Enron-flavoured layer: energy, gas, power, california, ees.

What employees threw away



Deletions are flat for years, then spike: $\sim 15,000$ in October 2001 and over **40%** of Feb / Jun 2002 mail trashed. Window matches the public collapse.

Open challenges

- **91.9%** of subjects are shared across multiple emails, weakening subject-based threading
- Body text needs cleaning before any NLP (forwarded blocks, attachment placeholders, quoted replies)
- Many files are duplicated across sender and recipient mailboxes, plus extra copies from forwarding
- Engineered header features are mostly uncorrelated so far
- Surfacing notable individual messages is still manual

Where we stand

The corpus is large, real, and unevenly distributed. Volume, folder shape, sender domains, activity rhythm, and vocabulary all line up with what we would expect from a real US energy trader.

The clearest single finding is the deletion spike in late 2001 to mid-2002, which sits exactly on top of Enron's public collapse. Thread reconstruction is the largest open problem and the focus for next week.

Questions?